Weekly correlations

Fourteen market variables, 2003–2026

DSC–AC research workflow

September 2026

Descriptive results and a bounded MATLAB estimation pilot

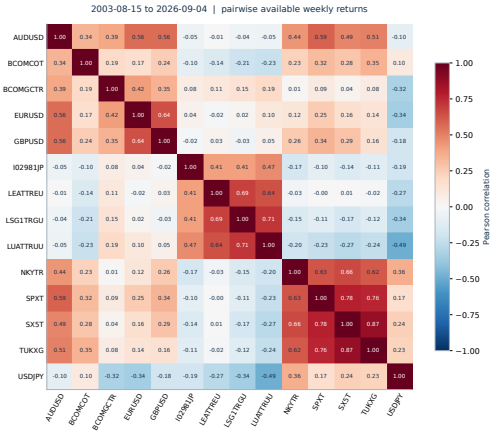
Research findings

- 1,204 weekly observations support 91 pairwise paths.
- Largest absolute full-sample correlation: TUKXG / SX5T (0.87).
- The 52-week SX5T / BCOMCOT path ranges from -0.55 to 0.83.
- ADF rejects a unit root for 14 return series. KPSS sensitivity qualifies the stationarity conclusion.

Use: historical comparison of the supplied returns. The current evidence does not establish trading performance or posterior convergence.

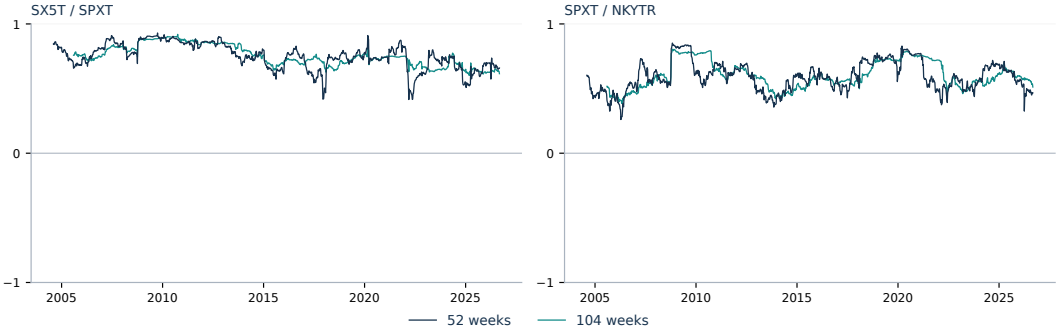
Source: supplied market levels; generated weekly-research outputs. Native quotes.

Full-sample correlations



Source: supplied market levels; generated weekly-research outputs. Native quotes.

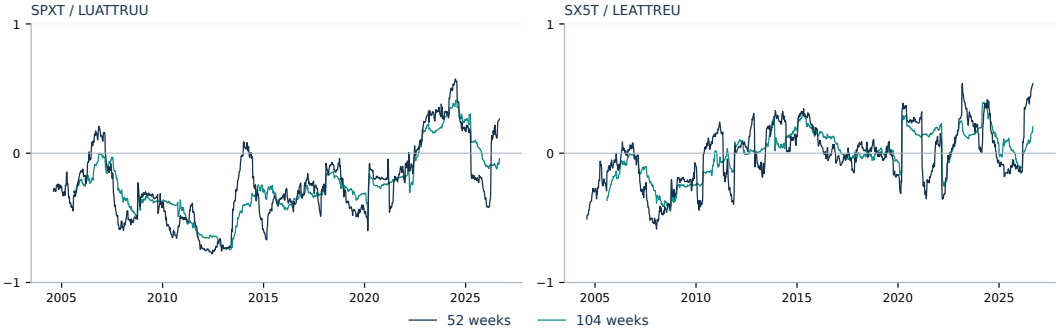
Selected equity-index pairs



Weekly native-quote returns. Shorter windows respond faster and fluctuate more.

Source: supplied market levels; generated weekly-research outputs. Native quotes.

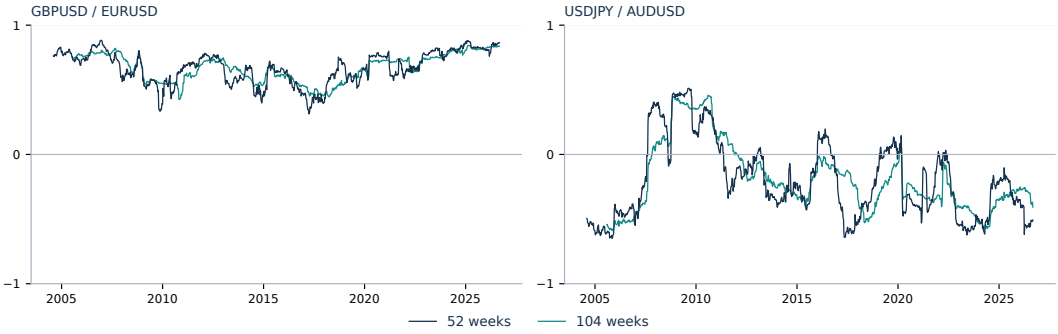
Selected cross-market pairs



These are descriptive return correlations. Pair labels retain the supplied identifiers.

Source: supplied market levels; generated weekly-research outputs. Native quotes.

Currency-pair correlations



USDJPY quotes yen per US dollar. EURUSD, GBPUSD and AUDUSD use US dollars per unit of the other currency. Quote direction affects correlation signs.

Source: supplied market levels; generated weekly-research outputs. Native quotes.

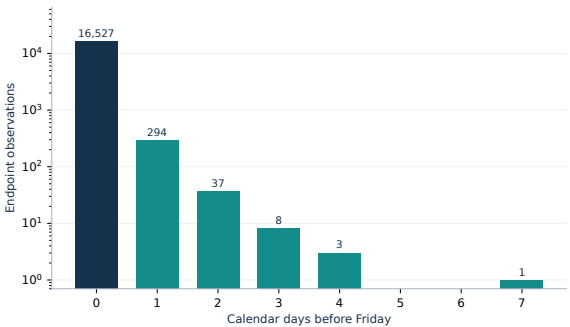
Correlation sensitivity to the window

Pair	Latest 52	Latest 104
SX5T / SPXT	0.66	0.62
SPXT / NKYTR	0.47	0.51
SPXT / LUATTRUU	0.26	-0.04
SX5T / LEATTREU	0.54	0.20
GBPUSD / EURUSD	0.86	0.84
USDJPY / AUDUSD	-0.51	-0.41

End date: 2026-09-04. Pearson correlations on fixed calendar windows. Overlapping windows make adjacent estimates dependent. The range of a rolling line does not measure estimation uncertainty.

Source: supplied market levels; generated weekly-research outputs. Native quotes.

Weekly alignment and the 2019 closure



Source: supplied market levels; generated weekly-research outputs. Native quotes.

1,204 returns across 14 series.

Friday endpoints use past observations only, with a four-day age limit.

One NKYTR endpoint is seven days old. The baseline flags the zero/catch-up returns. An optional sensitivity mask preserves all weeks.

JPX, 15 January 2019 holiday notice. The unfinished final week is excluded.

Stationarity and serial dependence

- Return ADF rejects a unit root for 14 of 14 variables.
- KPSS at 13 weeks flags: I02981JP, LEATTREU, LSG1TRGU.
- Lag and bandwidth sensitivity matter for mean specification.
- Serial-dependence tests motivate an AR-mean sensitivity before production.

ADF tests a unit-root null. KPSS tests a stationarity null. Neither test establishes a stable unconditional covariance.

Source: supplied market levels; generated weekly-research outputs. Native quotes.

The joint Bayesian target

$$y_t = B_t + \varepsilon_t, \qquad \varepsilon_t \sim N(0, D_t P_t D_t),$$
$$r_t = \mathrm{vecl}(\log P_t), \qquad r_t = r_{t-1} + v_t.$$

- 14 time-varying means and 14 log variances.
- 91 correlation coordinates produce positive-definite matrices.
- Historical smoothing uses the entire sample.
- $p = 0$ baseline. Serial dependence remains a production design issue.

Archakov and Hansen (2021). Actual entries of P_t, rather than matrix-log coordinates, are the reported correlations.

Bounded sampler pilot

Pilot quantity	Recorded value
Completed iterations	20
Retained posterior draws	0
Elapsed wall time	10.83 minutes
Mean time per completed iteration	32.46 seconds
Stop status	Iteration cap reached
Correlation / volatility proposals	7,809 / 1,419
Peak MATLAB process RSS	1.149 GiB

Whole sweeps: 4.32 times faster over 13 common sweeps; fixed-input kernel: 4.31 times faster.

Pilot draws support numerical and timing checks only. They do not support posterior credible intervals or convergence claims.

Production decision

Recommendation: hold production.

Acceleration is implemented; mean/prior sensitivity and convergence assessment remain.

Illustrative 4,000-iteration chain: **36.1 hours** at pilot speed. Four serial chains: **144.3 hours**. Projections exclude checkpoint writes and production posterior-chunk storage. No convergence guarantee.

- Four independent chains with rank-normalized $\widehat{R} < 1.01$.
- Bulk and tail ESS of at least 400 for reported quantities.
- Review mean dynamics, prior-window sensitivity and closure treatment.

Vehtari et al. (2021). A separate decision is required before a multi-day production run.

Technical appendix: priors and storage

First 104 returns calibrate initial priors, with 5% covariance shrinkage. All observations remain in the likelihood.

The 104-week rolling calibration sets inverse-gamma innovation priors with shape 10. Sensitivity windows are 52 and 156 weeks. Full-sample calibration reuses data.

Cumulative-normal random-walk draws replace dense temporal factors exactly. Checkpoints preserve chain identity, RNG state, dates and the source hash.

MAT-v7.3 chunks store distinct correlation entries. Per-chain burn-in exclusion occurs before summaries.

Technical appendix: diagnostic detail

Dependence diagnostic	Rejections / tests
ARCH	14 / 14
LjungBoxReturns	5 / 14
LjungBoxSquaredReturns	14 / 14

These are nominal 5% decisions without multiplicity correction. The full report records stationarity specifications and all 91 correlation paths.

Sources and reproducible outputs

- Supplied market data: 4 August 2003 onward, excluding USDCNH.
- Arias, Rubio-Ramírez and Shin (2023), dynamic stochastic-correlation models.
- Archakov and Hansen (2021), *A New Parametrization of Correlation Matrices*.
- Murray, Adams and MacKay (2010), *Elliptical Slice Sampling*.
- Vehtari et al. (2021), rank-normalized MCMC diagnostics.
- MathWorks, `adftest` and `kpsstest` documentation.
- Japan Exchange Group, 15 January 2019 holiday notice.

The companion report contains equations, complete citations and all 91 pairwise charts. CSV/MAT files and editable LaTeX sources accompany both PDFs.